
MARK JEFFERY AND JAMES ANFIELD

B&K Distributors: Calculating Return on Investment for a Web-Based Customer Portal

Jim Anfield, vice president of business development for JDA Consulting, originally learned about B&K Distributors from an acquaintance, Nancy O'Neil, B&K Distributors's vice president of sales. From his initial conversations with O'Neil, Anfield learned that B&K was trying to enhance its relationship with one of its key strategic accounts. However, B&K was reluctant to add additional customer service and sales staff to build this business because of a cost-cutting program that was in place. O'Neil also mentioned in passing how she wished B&K could somehow automate some of the paper pushing it did to service this account: "There are just so many faxes and phone calls to deal with, and I know that other companies have successfully dealt with these issues using Internet technology!" exclaimed O'Neil.

Anfield knew that his firm could be a tremendous resource to B&K. As a ten-year veteran of the information technology (IT) industry, he had helped numerous organizations in the distribution industry leverage technology to achieve both productivity and cost improvements as well as drive top-line growth. Anfield was convinced that his firm could assist B&K and add significant value.

After his initial meetings and conversations with O'Neil, Anfield set up a meeting with B&K's CEO, Jonathan Rodriguez, and the B&K senior management team. In this meeting, they discussed B&K's challenges with this key strategic account and potential strategies to address the various issues that B&K was facing. After about an hour of discussion, Anfield felt that the time was right to get specific regarding a potential strategic marketing and technology solution that would meet B&K's needs.

"Have you thought about using an Internet portal combined with a targeted marketing and communication program? This portal would drive incremental sales penetration of your strategic accounts and, in the process, offer better customer service while reducing your processing costs," Anfield said.

"I haven't been sold on these new technologies," replied Rodriguez. "Look at all those dot-coms that failed to generate meaningful sales, let alone profits."

"What if I could show you that you could realize a substantial financial return on a Web investment and even recover your costs within two years of the launch?" Anfield responded.

Rodriguez thought for a moment and then replied, “If the numbers work out, I would be happy to consider such a strategic investment. Why don’t you and Nancy O’Neil work on developing a business case for this project?”

For the next several weeks, Anfield and O’Neil spent time talking with people in different functional areas throughout B&K. Their objective was to help build the cost and revenue assumptions and projections for the strategic marketing initiative. They then began to build a set of “what-if” scenarios to illustrate the return B&K could expect to earn on its investment. As part of the process, they reviewed all of the assumptions with B&K senior management to ensure that they would be comfortable with the final numbers.

Once the business discovery phase was completed, Anfield needed to distill all of the information and calculate the projected return on investment (ROI). He wondered if the project would have a positive ROI and if Rodriguez would approve the initiative.

B&K Distributors

B&K Distributors was a \$100 million distributor of paper and packaging products based in Chicago, Illinois. The company focused on serving the specialized packaging and related needs of restaurants and institutional food-service businesses throughout the United States. The company’s strategy was to work closely with its customers to develop a tailored approach to meeting all of a particular customer’s specialty packing needs.

The majority of the company’s orders flowed through its central warehouse facility in Chicago. The company utilized common carriers to transport its products to customers and a dedicated in-house sales force to sell its products.

While a significant portion of the company’s sales came from small to medium-sized businesses located throughout the central United States, B&K’s management had also built a strong relationship with one of the nation’s leading fast-food restaurant chains. The company viewed this relationship as key to its continued growth and prosperity.

The majority of the company’s customers, including a significant set of the key account fast-food franchises, traditionally had little technology experience—few used the Internet or other technology tools in their day-to-day business operations. Even so, the franchises’ use of the Internet—accessing the Web and e-mail in particular—appeared to be on the rise in recent years, following the trend of typical American household users of the Internet.

The Preferred Supplier Agreement

Background

In 1997 the company signed a preferred supplier agreement with a leading fast-food restaurant chain and its nationwide network of franchisees. While franchisees were not required to purchase their supplies from B&K, the contract enhanced the visibility of B&K’s products to the franchisees. For example, B&K’s line of products was included in the corporate sourcebook

delivered to franchisees. Further, shipping and handling were improved by this preferred supplier agreement.

While the preferred supplier agreement was advantageous for B&K (it earned a fair profit on this business), the company had plateaued in its efforts to fully penetrate the account base and was unable to gain additional business from a significant percentage of the independent franchisees. The franchise market was saturated, and the growth rate was expected to be 2 percent for the next few years. This estimate was the net percent of new franchise restaurants that were projected to open each year.

Based on a review of order information provided by B&K, Anfield learned that the company did business with approximately 50 percent of the 3,400 independent franchise locations. One of the most significant reasons for this lack of penetration was that B&K had limited resources and was able to assign only one salesperson to service the entire nationwide franchise network.

A, B, and C Tier Customers

After a few years of experience, the B&K sales and marketing department noticed a very distinct segmentation of the franchises. Specifically, the company recognized that some of the franchises were very loyal customers—ordering very frequently and predictably—whereas other customers tended to order less frequently, treating B&K as a secondary supplier instead of their primary source.

After reviewing data from the company's sales order systems, the marketing and sales department segmented the franchise business into three tiers of customers based on the frequency of their orders. The company classified these customers as A, B, and C tier customers. **Exhibit 1** describes the ordering behavior of the three tiers of customers.

O'Neil noted that B&K faced the "80/20 rule" in terms of its customer tier structure, where 80 percent of revenue derived from 20 percent of the customer base. She told Anfield:

Getting customers to A or even B status enhances our revenue and profitability prospects significantly. Those A customers mean so much to the company. Unfortunately, if we simply keep our sales and marketing approach status quo, I wouldn't expect any change in the composition of our A, B, and C accounts. Further, without any additional marketing focus, I don't see our overall penetration of the franchise system itself changing—well, maybe we could gain a half of a percent or so a year the way things are now.

Economics of the Relationship

Anfield also gathered information about the existing profitability of the relationship by speaking with B&K's sales and marketing staff. Information regarding the profitability of the agreement is summarized in **Exhibit 2**.

The sales and marketing department noted that, in accordance with the account tier structure, without any change in the marketing plan, B&K would realize only annual inflation increases in

its average order size. B&K management also expected both the cost of products and the fixed cost per order to rise based on inflation for the foreseeable future.

B&K Departments

Customer Service/Order Entry Department

B&K employed approximately fifteen customer service representatives to handle incoming sales orders. A sales order could be received in one of three ways:

- 1-800 phone orders from customer or sales representative
- Fax orders from customer or sales representative
- Mail orders from customer only

B&K's order entry department took mail and fax orders and keyed them into the company's order entry application. Phone orders were keyed into the order entry application at the time the customer service representative fielded the customer's or sales representative's phone call.

The costs of both key account and field sales calls were quite high. As part of an earlier cost accounting study, the company determined the fixed cost it took to process each customer order. **Exhibit 3** summarizes the information gathered by the company regarding these costs.

IT Department

At the time, B&K had little in the way of an online presence. This was partially a result of not employing any information system personnel trained in Web-specific skills.

The information system department at B&K consisted of approximately ten individuals whose main duties were focused on two specific areas:

- Maintenance of the company's local area network and e-mail system, as well as overall end user support
- Maintenance, support, and troubleshooting of the client's back-office applications, including the following:
 - Inventory management
 - Accounting systems (general ledger, accounts receivable, accounts payable, payroll, and related systems)
 - Order entry and management systems
 - Plant and warehouse operations

Based on the limited number of resources and other priorities of the information systems staff, Anfield felt that outsourcing ongoing support and maintenance for the Web portal initiative might be the best option for B&K.

The Web Portal Business Case

Overview

After approximately a month of gathering detailed information about B&K, Anfield and his firm's team were ready to help build the business case for the strategic marketing campaign and Web portal system. Based on his previous experience with other clients, as well as his discussions with employees of B&K, he felt that a targeted marketing program with a customer-tailored Internet portal as its centerpiece was a way in which B&K could help build both its revenues and profits with the strategic account.

To help illustrate to B&K what the concept might look like, Anfield had his technical staff mock up an HTML-based prototype of the portal, which is shown in **Exhibit 4**.

The portal concept would offer a significant number of features, including:

- Customer service response application
- Shopping cart functionality
- Order entry module
- Order tracking status feature

Anfield knew that the overall strategic program could bring a number of benefits to an organization that had historically relied on live sales contact and phone or fax customer service/order entry. Specifically, Anfield and B&K's sales and marketing department identified the following items as the major sources of potential improvement:

- Additional penetration of the customer base
- Enhanced customer order frequency
- Increased average order size
- Reduction in the transaction cost per order

With O'Neil's assistance, Anfield was ready to quantify the benefits that could be attained from each of the above items. From discussions with O'Neil, Anfield discovered that B&K's cost of capital was 12 percent and its tax rate was 38 percent.

Increased Penetration

While B&K management expected relatively little additional franchise penetration by following the current marketing plan, they felt the self-service Web portal had the potential to greatly enhance the company's visibility to franchisees that they did not already serve. In addition, a self-service Web portal was especially important for franchisees located far away from B&K's physical location, as it was very difficult for B&K, with only one salesperson dedicated to the account, to visit or contact these distant locations.

As part of the research for this project, B&K's sales and marketing department reviewed industry case studies and conducted both primary and secondary market research. It concluded that penetration could rise significantly, up to a sustained overall 70 percent penetration rate within four years of implementation. **Exhibit 5** illustrates B&K management's estimates of franchise penetration over the next several years.

Enhanced Order Frequency

Based on the primary and secondary market research cited earlier, B&K's sales and marketing department also believed that it could change the purchasing behaviors of not only its existing customer base but also the new customers reached by this e-business initiative. The existing A tier customers were considered to be already purchasing their full product needs from B&K. However, O'Neil's team estimated that significant benefits could be realized with the B and C tier customers.

The focus, therefore, would be to turn B customers into A customers and to turn C customers into B customers. The net result would be a much larger percentage of A and B customers than before. Quantitatively, this transfer of customers between segments is represented in **Exhibit 6**.

Increased Order Size

Through its current channels, B&K's average order size from each franchisee was \$250, and without changes to the marketing approach, this number was expected to grow only with inflation. With the combination of the targeted marketing campaign and the Web portal solution, B&K's marketing and sales department believed that order size would grow substantially, as the company would be able to offer a much larger number of stock keeping units (SKUs) to the franchise network for essentially zero marginal cost.

B&K's entire product catalog would be included on the Web site, in contrast to the limited product lists provided in the franchisee catalog or maintained by the B&K salespersons. Overall, B&K management felt that it could double the increase in the average order size to 6 percent per year versus the 3 percent inflationary increase. The growth in order size is summarized in **Exhibit 7**.

Reduced Costs

In addition to building revenues through the various mechanisms described, another appealing feature of the proposed initiative was the ability of a Web portal solution to significantly reduce B&K's fixed costs for processing an order.

Anfield proposed integrating the Web front-end ordering screens with the company's existing order management application. With such integration, any Web order would not require human interaction to be processed—the data would be immediately transferred to the company's order management system.

With relatively limited interaction, the fixed cost to process a Web order would be almost zero, compared to transaction costs in the high \$20 range for the company's other channels.

Conservatively, Anfield estimated the cost to process an order using the Web portal to be approximately \$3 per order. This would cover items such as system monitoring and disk storage space. See **Exhibit 8** for cost for each order by channel. A conservative inflation estimate for the Web cost was 1 percent per year, and a 3 percent inflation rate was assumed for the phone and fax costs.

After the new system was in place, B&K would try to drive traffic and orders to the Web site, as opposed to the traditional order channels. The company's targeted marketing campaign would highlight the ease of use and offer incentives for using the site (Web specials, etc.). The company even considered allotting some promotional funds to train and coach franchisees one-on-one in the use of the application.

Based on these communication methods, the company hoped to encourage migration to the Web for ordering. **Exhibit 9** describes the projections that management felt were attainable based on similar experiences of other business-to-business organizations.

Web Portal Implementation Costs

It was about five and a half weeks into the information-gathering process when Anfield's phone rang. It was Rodriguez. "I like some of the revenue enhancement numbers I am hearing from people around here," remarked the CEO, "but Jim, just what is this thing going to cost me? I'm worried about the costs and how long it will take to get the system up and running. Also, my IT shop is small. How will we support this system after it is built?"

Anfield was aware of the need to make the system cost-efficient, both in terms of development and in ongoing support and maintenance. IT industry trade magazines were full of stories of projects long delayed and significantly over budget.

Anfield knew he needed to be careful and conservative with his cost estimates. He was also considering what type of contingency factor he might have to include in the projections to ensure that he could set management at ease regarding the cost projections.

Implementation

With the help of his consulting team, Anfield went to work scoping out a projected budget for the implementation of the project. Based on his work with similar companies, Anfield knew that a variety of skill sets would be needed to assist the client's cross-functional team with not only the technical development portion of the project but also the strategic and tactical marketing plans.

Anfield defined the team members and requisite skill sets for the project as follows:

- *Project Manager.* Provides day-to-day management of both the technical and operational components of the project.
- *Marketing Strategist.* Formulates, with the client's marketing team, the overall strategy and tactics to build revenue and enhance traffic; conducts audience/customer research and develops targeted messages as well as Web content; assists in executing plans and reviewing results.

- *Graphic Designer.* Formulates and executes the artistic vision of the Web site and related campaign materials.
- *Information Architect.* Develops the overall structure of the Web site and determines Web layout issues such as navigation and page layout.
- *Software Developer, Engineer, and Technical Architect.* Provides technical support to personnel who execute the overall plans and vision of the company; integrates the solution with the company's currently existing systems; develops computer code for needed functionality of the site.

Based on the proposed focus of the campaign, Anfield estimated the development project would last approximately four months. The project would begin with an initial marketing strategy visioning session, followed by the development of both the marketing plan as well as the Web-based solution. **Exhibit 10** describes the proposed budget for the development of the solution while **Exhibit 11** provides the phases included in the project.

The company planned to have O'Neil take responsibility for the project. She would also select a set of team members from B&K to assist in both the technical and operational portions of the project.

Ongoing Support and "Evolution" Costs

B&K also needed to consider the costs of supporting the application after implementation. Support costs would fall into three main categories:

- Web hosting costs
- Campaign continuance costs
- Additional consulting costs

WEB HOSTING

The most significant ongoing costs B&K could expect were the costs for a Web host provider. Anfield anticipated that B&K could negotiate a contract with an outsourced Web host for \$10,000 per month. This host would offer redundant processing systems and integration assistance, as well as storage and disaster recovery services.

As the company's current applications used a Microsoft/Intel infrastructure, Anfield recommended that B&K use a Microsoft/Intel-based infrastructure for the Web system as well. The portal platform would therefore use Microsoft SQL databases and related server software.

The outsourced Web host model was especially attractive to B&K because all the hardware and services ultimately needed to operate the system would be provided as part of the outsource contract. B&K would incur only a minimal amount of internal hardware and software costs to actually run the system—the B&K IT-related cost was so small that it was assumed to be zero for purposes of the ROI computation.

Exhibit 12 illustrates the infrastructure for a typical Web host specializing in small to medium-sized business applications. B&K's Web portal application would share this infrastructure with other companies that also purchased services from the Web host.

CAMPAIGN CONTINUANCE COSTS

The company also expected to spend a moderate amount on continuing marketing efforts in subsequent years. As Anfield informed B&K, the Web platform would allow the company to analyze hit rates and visitor travel paths, which would then help the company to continually tailor its message. These costs would include items such as additional promotional efforts and direct-mail campaigns.

WEB SITE MAINTENANCE

Finally, the company could expect to expend additional recurring costs on consultants to assist B&K in revising the Web site to better meet customer needs and to keep the Web site up to date. **Exhibit 13** describes the total projected annual support costs for the application.

The Proposal Review

In a week, Anfield and O'Neil were scheduled to present the results of their work to B&K senior management in the final proposal review meeting. Anfield was eager to learn what the ultimate ROI of this project would be, based on all the information they had gathered from B&K.

Anfield talked with O'Neil often, and she told him:

Jonathan [the CEO] is not a big believer in technology. I feel good about this idea, but I'm just not sure if Jonathan will ultimately approve [the project] based on the projected costs and benefits. We need to synthesize our business discovery and put together a realistic proposal if we hope to win final approval.

Anfield needed his team's assistance to prepare a thorough ROI analysis for the senior management review. He gave his team ROI spreadsheet templates that were based on previous consulting engagements (see **Exhibits 14A, 14B, 14C, and 14D**). The team could use these as a starting point for its analysis. Anfield wondered what the forecasted ROI for the Web portal would be, and if B&K's CEO, Jonathan Rodriguez, would approve the project.



Exhibit 1: A, B, and C Tier Order Data (Current State)

	Customer Tier		
	A	B	C
Percentage of accounts	25%	55%	20%
Number of orders per year	12	8	3

Exhibit 2: Revenue and Cost Data Per Order (Current State)

Average order size (equal across all three tiers)	\$250.00
COGS (cost of goods sold) percent	68.70%

Exhibit 3: Fixed Transaction Cost Per Order by Channel (Current State)

Channel	Percentage	Cost per Order
Phone	60%	\$32.50
Fax/mail	40%	\$25.00
Web	N/A	N/A
<i>Weighted average</i>	100%	\$29.50

Exhibit 4: Web Portal Prototype

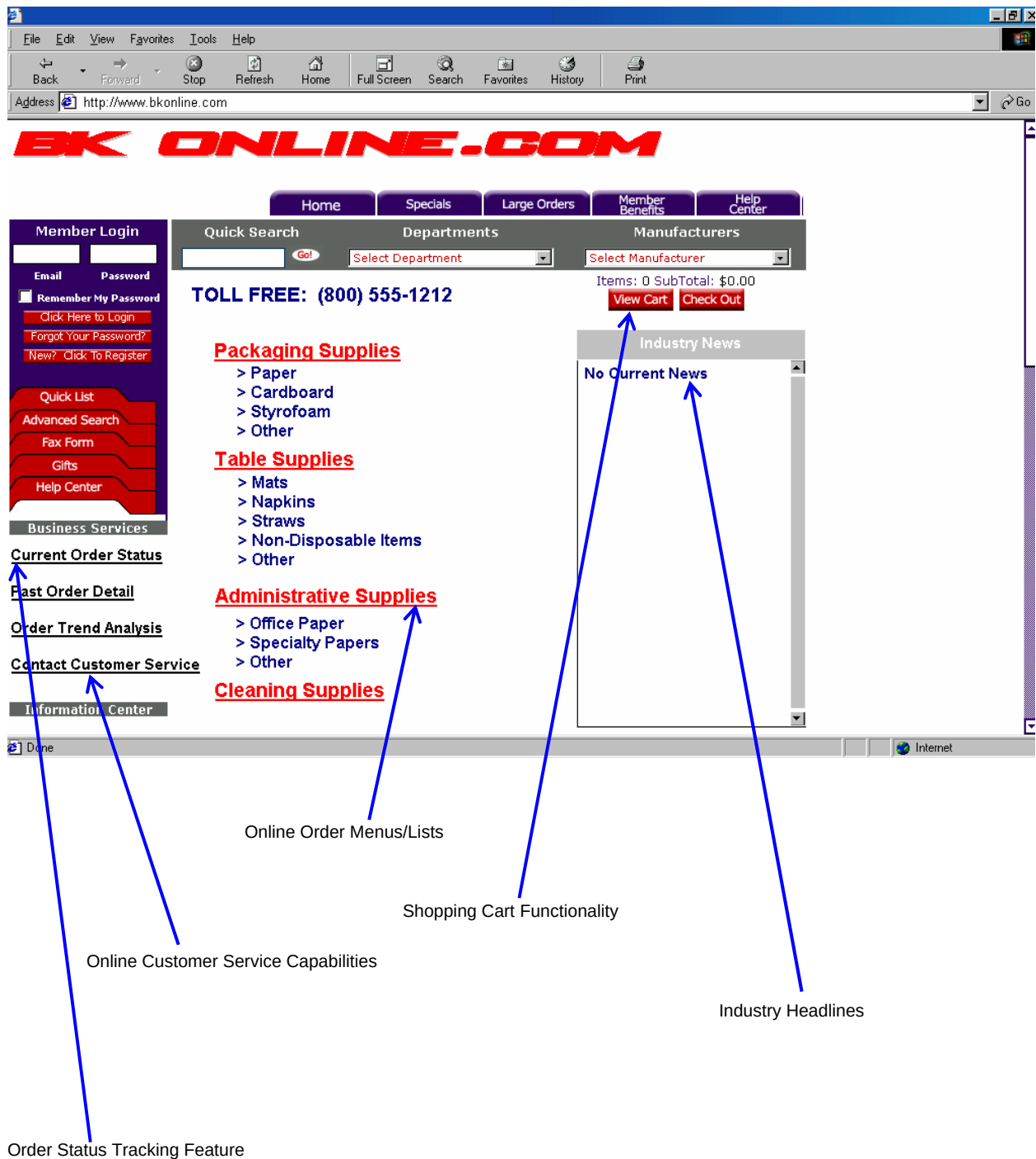


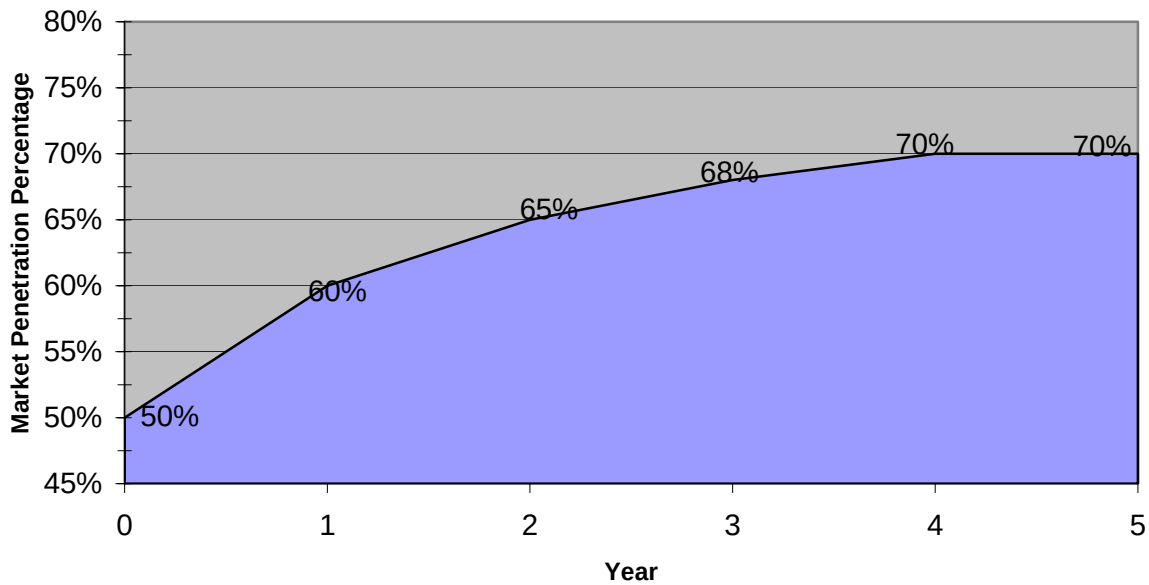
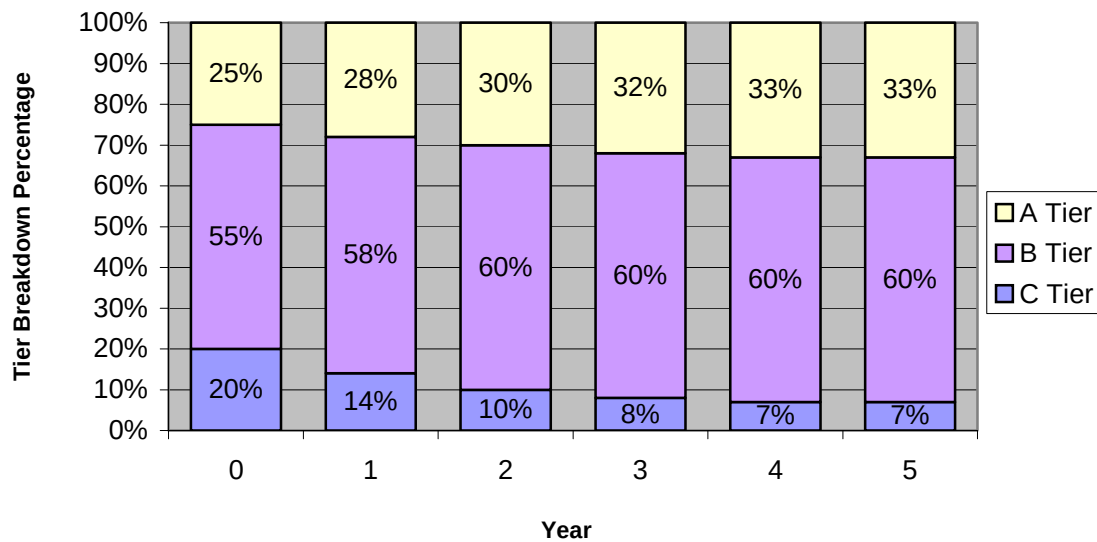
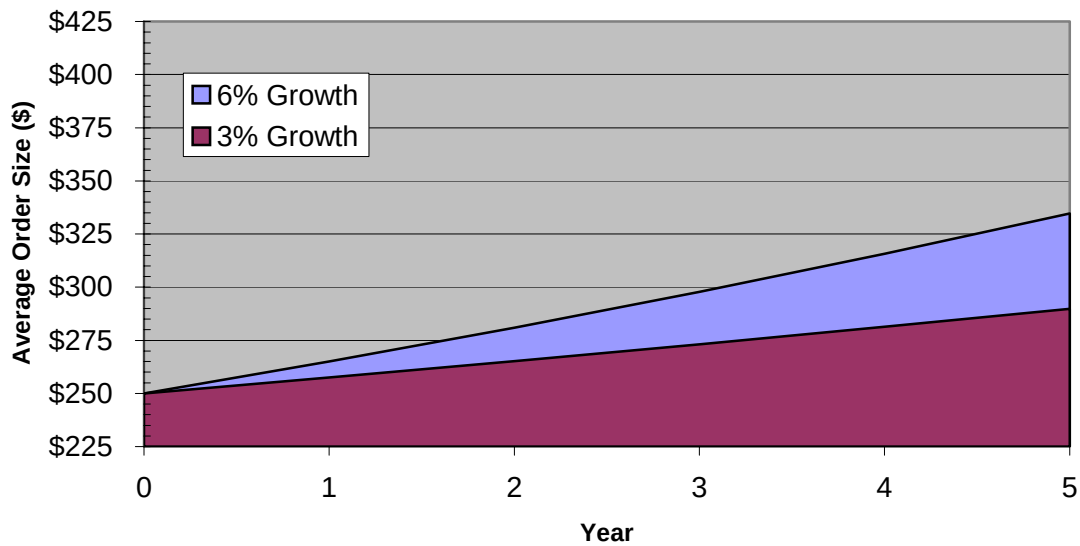
Exhibit 5: Franchise Penetration Over Time**Exhibit 6: Projected Tier Migration Over Time**

Exhibit 7: Order Size Improvement Over Time**Exhibit 8: Cost Per Order By Channel**

	Phone	Fax / Mail	Web
Cost per order	\$32.50	\$25.00	\$3.00
Inflation rate	3.0%	3.0%	1.0%

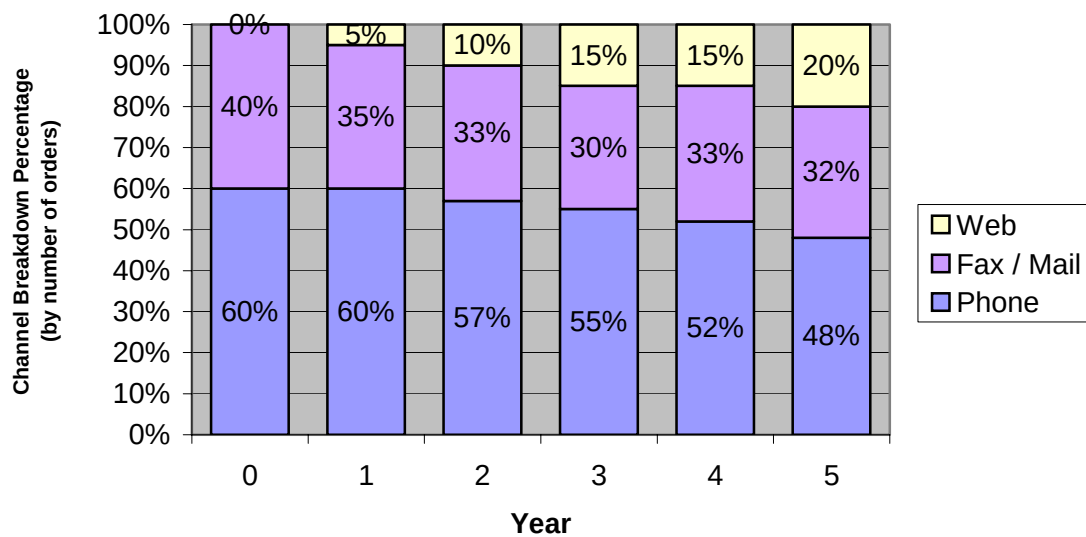
Exhibit 9: Order Channel Migration Over Time

Exhibit 10: Implementation Costs

	Full Time Equivalent (FTE) Percentage Per Month					
	Rate / Hr	Month 1	Month 2	Month 3	Month 4	
Project manager	\$200	25%	75%	75%	50%	
Marketing strategist	\$300	75%	20%	5%	5%	
Graphic designer	\$150	15%	75%	25%	10%	
Information architect	\$200	50%	75%	25%	7%	
Software developer	\$150	10%	50%	50%	50%	
Software engineer	\$150	10%	50%	50%	50%	
Technical architect	\$175	10%	100%	25%	5%	
					Total	
Project manager		8,000	24,000	24,000	16,000	72,000
Marketing strategist		36,000	9,600	2,400	2,400	50,400
Graphic designer		3,600	18,000	6,000	2,400	30,000
Information architect		16,000	24,000	8,000	2,240	50,240
Software developer		2,400	12,000	12,000	12,000	38,400
Software engineer		2,400	12,000	12,000	12,000	38,400
Technical architect		2,800	28,000	7,000	1,400	39,200
Total consulting costs		\$ 71,200	\$ 127,600	\$ 71,400	\$ 48,440	\$ 318,640
			Rate / Hour	# of Hours		
Outside specialists			200	40		\$ 8,000
			Cost / Year	Percentage		
Web hosting integration costs (as a percentage of first-year hosting costs)			120,000	50%		\$ 60,000
Grand total						\$ 386,640

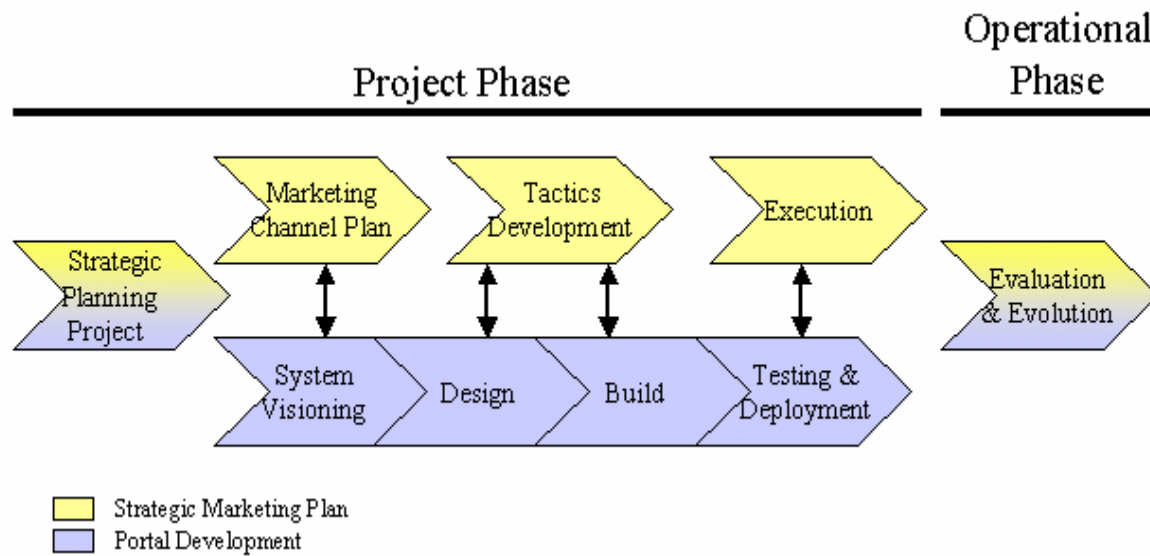
Exhibit 11: Project Phases

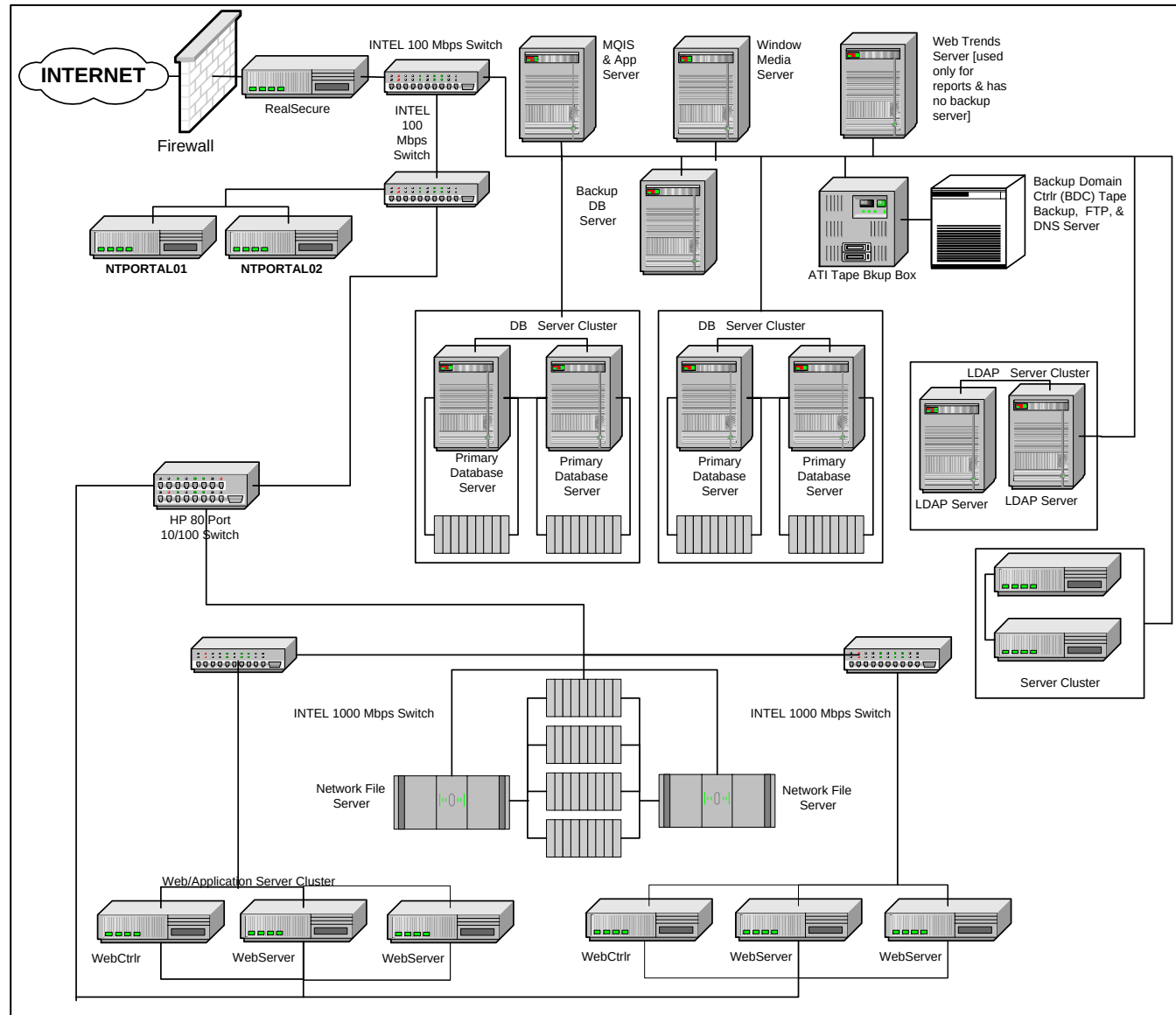
Exhibit 12: Web Host Infrastructure

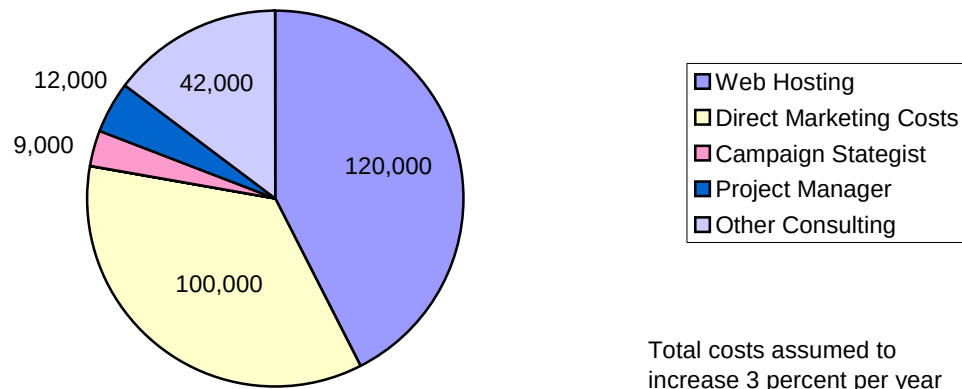
Exhibit 13: Annual Support Costs (\$)

Exhibit 14A: Return on Investment Template—Summary Page

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue						
Upside						
Baseline						
Incremental						
Net income						
Upside						
Baseline						
Incremental						
Incremental investment program (after tax)						
Upfront costs						
Ongoing costs						
Total						
Total (after tax)						
Total incremental cash flow						
Internal rate of return						
Net present value						
Discount rate						
Tax rate						
Support cost inflation rate						

Exhibit 14B: Return on Investment Template—Upside Projection

(Numbers in thousands)	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Number of franchise locations						
Growth						
Penetration						
Number of customers						
Tier structure						
Percent						
A # of locations						
Order frequency						
Percent						
B # of locations						
Order frequency						
Percent						
C # of locations						
Order frequency						
Average order size						
Growth						
Total transactions						
A accounts						
B accounts						
C accounts						
Total						
Revenue						
A accounts						
B accounts						
C accounts						
Total						
Unit margin						
COGS %						
Gross margin per order						
Processing cost per order						
Net contribution per order						
Account contribution						
Transactions						
Net margin per order						
Operating income						
Tax						
Tax rate						
Tax						
Net contribution after tax						

Exhibit 14C: Return on Investment Template—Base Case Projection

(Numbers in thousands)	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Number of franchise locations						
Growth						
Penetration						
Number of customers						
Tier structure						
Percent						
A # of locations						
Order frequency						
Percent						
B # of locations						
Order frequency						
Percent						
C # of locations						
Order frequency						
Average order size						
Growth						
Total transactions						
A accounts						
B accounts						
C accounts						
Total						
Revenue						
A accounts						
B accounts						
C accounts						
Total						
Unit margin						
COGS %						
Gross margin per order						
Processing cost per order						
Net contribution per order						
Account contribution						
Transactions						
Net margin per order						
Operating income						
Tax						
Tax rate						
Tax						
Net contribution after tax						

Exhibit 14D: Return on Investment Template—Average Transaction Cost Matrix

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Percentage of orders by channel:						
Phone						
Fax/mail						
Web						
Total percent						
Cost per order (\$):						
Phone						
Fax/mail						
Web						
Weighted average cost						
WAC without Web						
Difference						
Inflation increases:						
Phone						
Fax/mail						
Web						